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FactoryFlow: A Mobile Platform for Intelligent Industrial Maintenance KPI Collection and Automated Reporting

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GENERAL INTRODUCTION

CHAPTER 1

INDUSTRIAL CONTEXT AND PROBLEM DEFINITION

I. Introduction

Digital transformation in industrial environments extends beyond the automation of production equipment. It also concerns the way operational information is collected, structured, validated, analysed, and transformed into useful knowledge. Maintenance activities generate numerous indicators required for equipment monitoring, resource-consumption follow-up, anomaly identification, and periodic reporting. The value of these indicators, however, depends directly on the reliability and traceability of the underlying data.

During the internship at **Alf Mabrouk Nutrition Animale**, an important limitation was identified in the maintenance reporting process. Several **Key Performance Indicator (KPI)** values were communicated through heterogeneous WhatsApp messages and later transferred manually into reporting files. Although this communication channel enabled rapid information exchange, the absence of a standardized acquisition process introduced additional interpretation and transcription work. The existing workflow is presented in **Section IV**.

The internship project therefore focused on transforming this fragmented process into a structured and traceable digital workflow. This led to the development of **FactoryFlow**, a mobile platform dedicated to intelligent KPI acquisition, human-controlled validation, centralized storage, automated reporting, and maintenance-oriented analysis. The project objectives are detailed in **Section VIII**.

A complementary web application dedicated to dosage deviation analysis was also developed to address another industrial need related to production monitoring. While **FactoryFlow** focuses on trustworthy maintenance information and decision support, the **Dosage Analysis Web Application** focuses on detecting and monitoring deviations between expected and measured production quantities. Its role within the internship is introduced in **Section X**.

This chapter presents the industrial context, the existing maintenance workflow, its limitations, the associated data-quality challenges, and the objectives that shaped the proposed solutions.

II. Host Organization and Industrial Context

Alf Mabrouk, also known as La Marocaine d'Aliments Composés, is a Moroccan company operating in the **animal nutrition sector**. Established in 2012 and located in the **industrial zone**

of **Bouznika**, the company specializes in the manufacture and commercialization of compound feed intended for livestock and poultry.

Alf Mabrouk belongs to the poultry-related activities of **Cap Holding**, a Moroccan industrial and investment group active in cereal-related activities, industrial milling, and agricultural raw-material trading. This affiliation places the company within a broader industrial ecosystem involving the transformation, storage, and management of agricultural raw materials.

From an operational perspective, the technical organization is structured around complementary functions including maintenance, production, and raw-material reception and transfer. These activities are supported by supervisors, team leaders, operators, and shift-based teams. **Figure 1.1** provides a simplified representation of this organization.

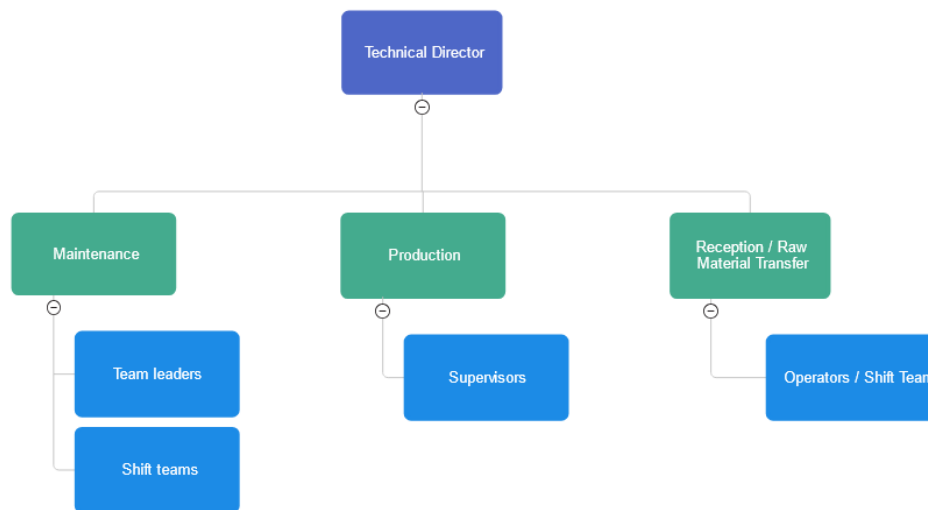


FIGURE 1.1 *Simplified technical organizational structure of Alf Mabrouk*

Production depends on the controlled transformation and combination of several raw materials and premixes. Each production recipe specifies the required ingredients and their target quantities.

The process begins with raw-material storage in dedicated silos. Ingredients are transferred to weighing systems according to the selected recipe before passing through milling. Smaller and more sensitive components are introduced through micro-dosing systems using precise weighing equipment. Liquid ingredients such as water and vegetable oils are also measured and injected before mixing. The resulting product may then pass through pressing lines, finished-product silos, and finally bulk handling or bagging. These principal stages are illustrated in **Figure 1.2**.

Simplified Process Overview

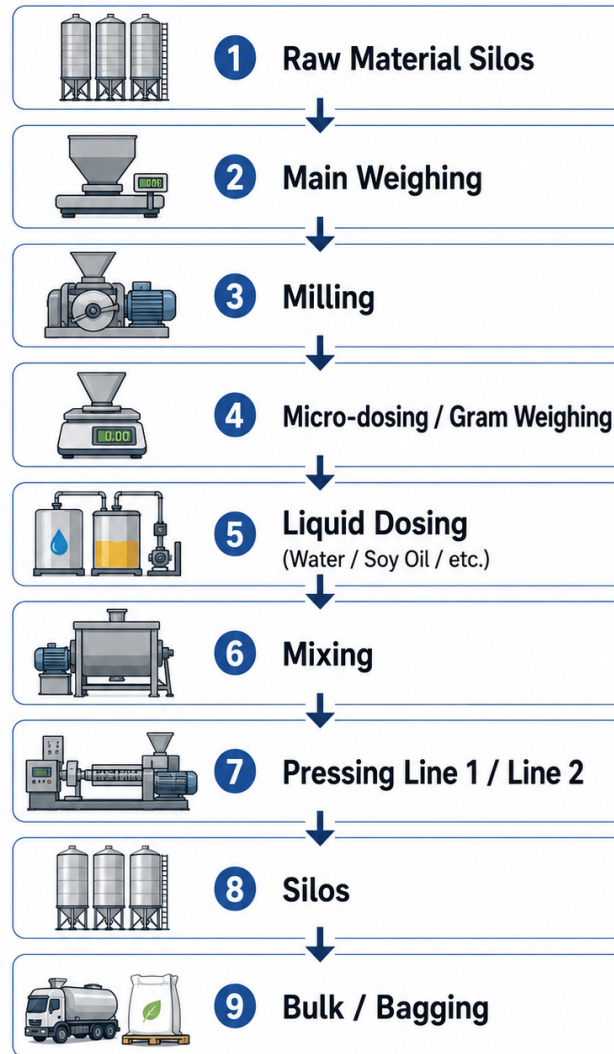


FIGURE 1.2 *Simplified overview of the compound-feed production process at Alf Mabrouk*

This production environment combines physical equipment with industrial information systems. Programmable logic controllers participate in equipment control and measurement, while recipes and operational information are managed digitally. Reliable industrial monitoring therefore depends not only on equipment operation, but also on the quality of the data collected from these activities. This relationship provides the context for the maintenance processes discussed in **Section III**.

III. Maintenance Context

Maintenance plays a central role in ensuring the availability and correct operation of the equipment involved throughout the production process. The stages shown in **Figure 1.2** depend on interconnected mechanical, electrical, electromechanical, and automated systems. A malfunction in one component may therefore influence one or several production stages.

Beyond direct interventions, the maintenance team follows operational indicators related to equipment utilisation and industrial utilities. These include, among others, operating hours, compressor measurements, water consumption, fuel quantities, oil levels, molasses levels, and other values required for technical monitoring.

As illustrated in **Figure 1.1**, maintenance activities involve different operational teams and shifts. Information collected in the field must consequently remain accessible and understandable regardless of its source or the employee who recorded it.

This context makes **data continuity and traceability** essential. A measurement may later be required for historical comparison, reporting, anomaly investigation, trend analysis, or technical decision support. The workflow previously used to transmit and structure these indicators is presented in **Section IV**.

IV. Existing KPI Collection Process

Before FactoryFlow, maintenance indicators were collected primarily through a combination of **WhatsApp communication and manual transcription**. Employees shared measurements collected during their activities or shifts through operational WhatsApp groups.

This approach was convenient for rapid human communication, but the messages were not designed for direct machine processing. Equivalent indicators could therefore appear with different spellings, abbreviations, separators, units, decimal formats, or ordering. The resulting data-quality challenges are discussed in **Section VI**.

A message could contain several indicators, only a small subset of them, or explicitly missing values. Once received, the useful information had to be manually identified, interpreted, and transferred into the reporting environment.

The original workflow is illustrated in **Figure 1.3**.



FIGURE 1.3 Existing maintenance KPI collection and reporting workflow

Although functional, this workflow introduced a manual transformation step between field information and the structured data used for reporting. Its principal limitations are analysed in **Section V**.

V. Limitations of the Existing Process

The existing process fulfilled the basic objective of transmitting operational information, but several limitations appeared once the messages had to be transformed into reusable data.

The first limitation concerned **manual effort**. Each message had to be interpreted before its values could be transferred into the reporting system.

A second limitation was the **risk of transcription and interpretation errors**. Decimal values could be misunderstood, information could be omitted, or a measurement could be associated with the wrong indicator.

A third limitation concerned **heterogeneous message formats**. Equivalent KPIs could be expressed using different labels, abbreviations, separators, units, and numerical representations. These variations are examined further in **Section VI**.

The process also had to preserve the distinction between **zero and missing information**. An unavailable measurement cannot safely be converted into zero because both states have different analytical meanings.

Finally, the workflow offered limited **source traceability**. Once a value had been manually copied into a reporting file, verifying its origin required returning to the original conversation.

The problem was therefore not WhatsApp itself, which remained a rapid operational communication tool. The need was to introduce a reliable processing layer between informal communication and structured industrial data. These limitations lead directly to the problem formulated in **Section VII**.

VI. Data Quality Challenges

The difficulty was not limited to transferring values from one interface to another. Before a measurement could be stored reliably, the system first had to determine **what the received information represented**.

The same KPI could appear with different capitalization, accents, abbreviations, punctuation, or spacing. Numerical values could use a comma or a period as a decimal separator, while certain formats could remain ambiguous without human confirmation. Units could be attached directly to values or written separately.

Another critical situation concerned **missing measurements**. Expressions such as dashes or explicit missing-value markers had to remain distinguishable from numerical zero. FactoryFlow therefore adopts a strict rule:

Missing information is not equivalent to zero.

Messages extracted from screenshots may also contain elements unrelated to maintenance KPIs, including names, timestamps, date separators, conversation headers, and interface metadata. These elements must be separated from business information while the original source remains preserved for traceability.

Some indicators may additionally contain **multiple related measurements**. A compressor entry, for example, may contain both a principal value and an associated percentage. Such information must remain semantically linked.

Reliable acquisition therefore requires normalization, classification, numerical interpretation, KPI matching, uncertainty management, and human validation. These constraints, combined with the limitations presented in **Section V**, motivate the project problem statement.

VII. Problem Statement

As illustrated in **Figure 1.3**, the maintenance reporting process relied on manual interpretation and transcription of heterogeneous operational information. The limitations identified in **Section V** and the data-quality constraints discussed in **Section VI** reduced standardization, traceability, and analytical exploitation.

The central problem can therefore be formulated as follows:

How can heterogeneous maintenance KPI information be transformed into reliable, structured, traceable, and analytically valuable industrial data while preserving human control over uncertain information?

The solution must reduce repetitive manual work while preserving the maintenance engineer's judgement. It should also move beyond data storage by transforming validated historical information into meaningful indicators, alerts, trends, and decision-support insights.

VIII. Internship Objectives

The main objective of the internship was to design and implement a mobile platform capable of transforming fragmented maintenance information into a structured, trustworthy, and exploitable industrial information flow.

More specifically, the project pursued the following objectives:

1. **Centralize maintenance KPI information** within a single mobile platform.
2. Support both **manual acquisition and image-based acquisition**, particularly from WhatsApp screenshots.
3. Use **Optical Character Recognition (OCR)** to extract textual information from imported images.
4. Process heterogeneous KPI messages through **deterministic parsing and normalization rules**.
5. Distinguish between **confirmed measurements, missing information, ambiguous values, noise, and unrecognized indicators**.
6. Preserve the original acquired content to ensure **end-to-end traceability**.
7. Maintain **human validation** before uncertain information enters the confirmed dataset.

8. Persist validated information within a **centralized database**.
9. Provide **historical consultation and interactive statistical analysis** based exclusively on validated data.
10. Introduce a **Maintenance Intelligence layer** capable of analysing historical KPI behaviour and identifying unusual patterns.
11. Apply **anomaly detection** to highlight measurements that deviate significantly from expected historical behaviour.
12. Provide **trend analysis and consumption forecasting** to support anticipation and technical decision-making.
13. Generate **contextual alerts and notifications** for abnormal measurements, relevant thresholds, missing information, and automated reporting events.
14. Generate structured **PDF and Excel reports** directly from validated maintenance information.
15. Support **scheduled report generation, email distribution, and document sharing**.

These objectives address the weaknesses identified in **Section V** while respecting the data-integrity constraints presented in **Section VI**.

IX. FactoryFlow Positioning

FactoryFlow is positioned between raw operational communication and reliable industrial decision support. It does not replace the communication habits of the maintenance team; instead, it provides a controlled environment in which information can be acquired, interpreted, validated, stored, analysed, and reported.

The platform combines **OCR-assisted acquisition, deterministic parsing, source-line classification, KPI matching, human validation, centralized persistence, historical analysis, automated reporting, and maintenance intelligence**. This architecture transforms informal operational information into a structured information asset.

A fundamental principle of the platform is:

Automation + Human Validation + Traceability = Trustworthy Industrial Reporting

Automation is therefore used where the evidence is sufficient, while ambiguous information remains under human control. Once validated, historical data becomes the foundation for statistics, anomaly detection, forecasting, and contextual alerts.

FactoryFlow consequently evolves beyond a conventional data-entry application. It acts as a **traceable industrial information pipeline** connecting field observations with reporting and maintenance-oriented decision support.

X. Complementary Dosage Analysis Project

In parallel with FactoryFlow, the **Dosage Analysis Web Application** was developed to address a production-related need concerning dosage deviations.

Production recipes define target quantities for the ingredients required by each batch. During execution, measured quantities can be compared with these targets and with the tolerance associated with each ingredient. The application centralizes these measurements and supports the detection, consultation, and follow-up of dosage anomalies.

The two projects therefore address complementary industrial problems. **FactoryFlow** focuses on transforming heterogeneous maintenance information into reliable and analytically exploitable data, whereas **Dosage Analysis** focuses on analysing deviations within the production process.

Their technical implementation and the relationship between both solutions are developed later in the dedicated Dosage Analysis chapter.

XI. Conclusion

This chapter established the industrial context and identified the weaknesses of the previous maintenance reporting workflow. Manual interpretation, heterogeneous formats, ambiguity, missing information, and limited traceability created the need for a more reliable information process.

These findings led to the development of **FactoryFlow**, combining assisted acquisition, deterministic interpretation, human validation, centralized persistence, automated reporting, and maintenance intelligence. The complementary Dosage Analysis project addresses a second industrial challenge related to production dosage deviations.

The next chapter translates these industrial needs into the **functional requirements, non-functional requirements, business rules, and system specifications** that guided the design of the proposed solutions.

CHAPTER 2

REQUIREMENTS ANALYSIS AND SYSTEM SPECIFICATION

I. Introduction

The industrial analysis presented in **Chapter 1** established the need for a structured process capable of transforming heterogeneous maintenance information into reliable and exploitable data. The next step was to translate this need into explicit system requirements and business constraints.

This chapter defines the expected behaviour of **FactoryFlow**, the actors interacting with the platform, its functional and non-functional requirements, and the business rules governing data interpretation and validation. Particular attention is given to **traceability, human-controlled validation, missing-value management, automated reporting, and maintenance-oriented intelligence**, as these elements directly respond to the limitations identified in **Section V**.

The chapter concludes with the acceptance criteria and the use case model that formalize the principal interactions between the user and the system.

II. Stakeholders and System Actors

FactoryFlow was designed for a limited group of maintenance professionals rather than for a large public user base. The principal system actor is therefore the **Maintenance Engineer**, who interacts directly with the mobile application throughout KPI acquisition, validation, consultation, analysis, and reporting.

The Maintenance Engineer acquires operational information, reviews interpreted measurements, resolves ambiguous or unrecognized values, confirms reports, consults historical data and statistics, and uses the analytical information produced by the platform.

Beyond the direct user, the information produced by **FactoryFlow** also supports the broader technical reporting process within **Alf Mabrouk**. Confirmed KPI histories, generated documents, alerts, and analytical results can therefore contribute to technical follow-up and decision-making.

Some operations are initiated automatically by the platform rather than directly by the user. These include **scheduled report generation, report distribution, notification delivery, anomaly analysis, and forecasting tasks**. These automated operations are considered when defining the system behaviour and use cases.

TABLE 2.1 Main FactoryFlow stakeholders and responsibilities

Stakeholder	Main responsibilities and interests
Maintenance Engineer	Acquires KPI information, reviews interpreted measurements, resolves uncertain data, confirms reports, consults historical indicators, and uses analytical results for maintenance follow-up.
Technical Management	Uses consolidated maintenance information, generated reports, statistics, alerts, and analytical indicators to support technical supervision and decision-making.
FactoryFlow Platform	Performs OCR-assisted acquisition, deterministic interpretation, validation support, centralized persistence, automated reporting, notifications, anomaly detection, forecasting, and maintenance-intelligence operations.

Summary of the principal actors involved in the FactoryFlow information workflow and their respective responsibilities.

The responsibilities summarized in **Table 2.1** provide the foundation for the functional requirements defined in **Section III**.

III. Functional Requirements

The functional requirements define the operations that FactoryFlow must provide in order to satisfy the objectives established in **Section VIII**. They cover the complete information lifecycle, from acquisition of raw maintenance data to validated reporting and maintenance-oriented analysis.

The requirements are organized according to the principal functional areas of the platform.

1. Authentication and secure access

FactoryFlow shall provide authenticated access to the platform and maintain the user session through a secure token-based mechanism. A user whose session is no longer valid must be informed clearly and redirected to authentication without losing the distinction between a session problem and an ordinary application error.

2. Maintenance KPI acquisition

The platform shall support several acquisition methods in order to remain compatible with the operational practices identified in **Section IV**.

The user shall be able to:

- 2.1. paste a maintenance message directly into the application;
- 2.2. import an image or WhatsApp screenshot;
- 2.3. extract visible text from an imported image using **OCR**;
- 2.4. enter measurements manually when no textual or visual source is available;
- 2.5. review the extracted or entered source before continuing to interpretation.

3. Deterministic interpretation of heterogeneous messages

The system shall transform acquired text into structured KPI candidates through deterministic processing. This processing shall support variations in capitalization, accents, punctuation, spacing, numerical separators, units, common aliases, and recognized OCR artefacts.

The interpretation process shall also distinguish between:

- 3.1. recognized KPI measurements;
- 3.2. ambiguous measurements requiring human review;
- 3.3. explicit missing values;
- 3.4. duplicate observations;
- 3.5. unrecognized KPI-like information;
- 3.6. source metadata and deterministic noise.

The distinction between these states is required by the data-quality constraints described in **Section VI**.

4. Human-controlled Review workflow

Before information becomes part of the confirmed dataset, FactoryFlow shall provide a Review stage in which every uncertain element has an explicit resolution path.

The platform shall allow the user to:

- 4.1. validate an ambiguous but acceptable measurement;
- 4.2. confirm a duplicate occurrence without merging it with another source observation;
- 4.3. correct a detected value before validation;
- 4.4. preserve an explicitly missing value as **Non renseigné**;
- 4.5. replace a missing value manually and validate the correction;
- 4.6. associate an unrecognized line with an existing KPI;

- 4.7. create and associate a genuinely new KPI without leaving the Review workflow;
- 4.8. ignore confirmed metadata or noise while preserving its raw source;
- 4.9. save incomplete work as a draft and resume its verification later.

Report confirmation shall remain unavailable while unresolved or review-required information is still present.

5. KPI catalog and association management

FactoryFlow shall maintain a reusable catalog of KPI definitions.

The system shall support:

- 5.1. exact matching against canonical KPI labels;
- 5.2. matching through approved aliases;
- 5.3. ranked fuzzy suggestions when no exact association is available;
- 5.4. visible confidence information for fuzzy suggestions;
- 5.5. manual selection of another KPI when the proposed association is unsuitable;
- 5.6. inline creation of a new indicator;
- 5.7. duplicate prevention when an equivalent KPI already exists;
- 5.8. explicit memorization of a user-approved association when requested.

A weak fuzzy suggestion shall never be treated as a definitive association and shall not prevent the user from creating a genuinely new indicator.

6. Traceability and persistence

Every confirmed report shall be stored in a centralized database together with the information required to understand its origin and validation history.

FactoryFlow shall preserve:

- 6.1. the original acquired source;
- 6.2. confirmed KPI measurements;
- 6.3. explicit missing values;
- 6.4. separately confirmed duplicate observations;
- 6.5. ignored source lines where traceability is required;
- 6.6. the resolution decisions performed during Review.

This requirement implements the traceability principle introduced in **Section III**.

7. Historical consultation and statistical analysis

The user shall be able to consult previously confirmed maintenance reports and analyse the evolution of recorded indicators over time.

The statistical module shall provide:

- 7.1. historical KPI consultation;
- 7.2. filtering by relevant period or indicator;
- 7.3. statistical summaries calculated from confirmed measurements;
- 7.4. interactive trend visualization;
- 7.5. consultation of the date and exact value of selected historical points.

Missing information shall be excluded from numerical calculations and shall never be converted into zero.

8. Maintenance Intelligence

FactoryFlow shall exploit validated historical information to provide a higher analytical layer dedicated to maintenance follow-up.

This layer shall support:

- 8.1. detection of abnormal KPI behaviour and significant deviations;
- 8.2. analysis of historical trends;
- 8.3. forecasting of suitable consumption or maintenance-related indicators when sufficient historical data is available;
- 8.4. presentation of analytical findings in a form understandable by the maintenance engineer;
- 8.5. contextual alerts when an abnormal or relevant condition is identified.

The intelligence layer shall operate on **validated historical data**, not directly on unreviewed OCR or parser output.

9. Automated reporting

The system shall generate structured reports from confirmed maintenance information.

FactoryFlow shall support:

- 9.1. generation of **PDF** reports;
- 9.2. generation of **Excel** reports;
- 9.3. daily, weekly, and monthly reporting periods;

- 9.4. consolidation of multiple confirmed source observations;
- 9.5. preservation of missing measurements as **Non renseigné**;
- 9.6. storage and later consultation of generated reports.

10. Scheduling, notification, and distribution

The platform shall reduce repetitive reporting operations through controlled automation. It shall allow:

- 10.1. creation of recurring report schedules;
- 10.2. automatic generation of scheduled reports;
- 10.3. email distribution of generated documents;
- 10.4. mobile sharing of reports through compatible Android applications;
- 10.5. notification of relevant reporting events;
- 10.6. notification of maintenance-intelligence events requiring user attention.

Together, these requirements define the functional perimeter of FactoryFlow. The quality constraints governing their execution are specified in **Section IV**, while the rules that protect the semantic integrity of maintenance data are formalized in **Section V**.

IV. Non-Functional Requirements

In addition to the functional capabilities defined in **Section III**, FactoryFlow must satisfy quality requirements related to reliability, security, usability, traceability, maintainability, performance, and analytical integrity.

The principal non-functional requirements are presented in **Table 2.2**.

TABLE 2.2 *Main non-functional requirements of FactoryFlow*

ID	Category	Requirement
NFR-01	Data Integrity	The system shall preserve the distinction between confirmed measurements, missing information, ambiguous values, duplicate observations, unresolved information, and ignored source metadata.

Table 2.2 – continued

ID	Category	Requirement
NFR-02	Reliability	Validation, association, draft saving, report confirmation, and source-line resolution shall produce deterministic results without silently losing previously acquired information.
NFR-03	Traceability	Information acquired from an image, pasted message, or manual entry shall remain traceable to its original source throughout interpretation, validation, storage, and reporting.
NFR-04	Usability	The mobile interface shall provide a clear workflow in which uncertain, missing, or unrecognized information exposes an understandable resolution action.
NFR-05	Performance	Navigation, KPI review, historical consultation, and statistical interaction shall remain responsive on the target Android devices, without unnecessary blocking operations on the user interface.
NFR-06	Security	Protected resources shall require authentication, while credentials, authentication tokens, and application secrets shall be handled securely and kept outside exposed application logic.
NFR-07	Operational Resilience	Temporary communication failures shall be reported clearly and shall not silently corrupt confirmed information. Incomplete work shall remain recoverable through the draft mechanism when applicable.
NFR-08	Maintainability	Acquisition, interpretation, validation, persistence, reporting, and analytical responsibilities shall remain sufficiently separated to allow individual components to evolve independently.

Table 2.2 – continued

ID	Category	Requirement
NFR-09	Extensibility	New KPI definitions, aliases, interpretation rules, analytical mechanisms, and notification behaviours shall be introducible without redesigning the complete platform.
NFR-10	Interoperability	The Android application, backend, PostgreSQL database, OCR runtime, email services, and generated documents shall communicate through clearly defined interfaces and standard data representations.
NFR-11	Deployment	Environment-dependent parameters such as API endpoints, database connections, OCR-service addresses, storage locations, and secrets shall be externalized and configurable according to the deployment environment.
NFR-12	Analytical Reliability	Statistics, anomaly detection, trend analysis, and forecasting shall operate on validated historical measurements. Missing information shall never be silently interpreted as numerical zero.

The requirements in **Table 2.2** reinforce the same principle established during the analysis of the existing process: automation must improve the handling of industrial information without reducing its reliability or traceability. In particular, analytical functions operate on validated historical data rather than directly on raw OCR or parser output.

These quality constraints are complemented by the business rules defined in **Section V**.

V. Business Rules

The business rules define the conditions that FactoryFlow must respect when interpreting, validating, storing, and analysing maintenance information. Unlike interface behaviour, these rules protect the semantic integrity of the data and therefore remain applicable throughout the complete processing workflow.

The principal business rules are presented in **Table 2.3**.

TABLE 2.3 Business rules governing FactoryFlow data processing

ID	Rule	Description
BR-01	Missing is not zero	An explicitly missing measurement shall remain represented as missing or Non renseigné . It shall never be converted automatically into the numerical value zero.
BR-02	Zero is a valid value	A source measurement explicitly equal to zero shall remain a numerical zero and shall not be interpreted as missing information.
BR-03	Deterministic interpretation	Normalization and KPI interpretation shall rely on deterministic rules. Equivalent capitalization, accents, spacing, punctuation, separators, recognized units, and known OCR artefacts may be normalized without changing the original source retained for traceability.
BR-04	Ambiguous numbers require review	When a numerical representation admits more than one credible interpretation, the system shall preserve the detected value and request human validation rather than silently selecting an interpretation.
BR-05	KPI matching priority	KPI association shall prioritize an exact canonical match, followed by an approved alias. Fuzzy matching may provide ranked suggestions only when no higher-authority match exists.
BR-06	Weak suggestions are non-authoritative	A weak fuzzy suggestion shall never be treated as a confirmed association and shall not prevent the user from selecting another KPI or creating a new indicator.
BR-07	No silent learning	A new alias or reusable association shall be memorized only after an explicit user decision. The system shall not silently learn from an uncertain or automatically proposed association.

Table 2.3 – continued

ID	Rule	Description
BR-08	Human validation of uncertainty	Ambiguous measurements, review-required warnings, corrected missing values, and unresolved KPI associations shall require an explicit user resolution before they enter the confirmed dataset.
BR-09	Duplicate observations remain traceable	Multiple occurrences of the same KPI in a source shall not be silently merged or overwritten. Each legitimate occurrence may be validated independently and shall retain its source traceability.
BR-10	Composite measurements remain linked	When a source line contains several semantically related measurements, such as a principal compressor value and an associated percentage, these values shall remain linked to the same source observation.
BR-11	Correcting a missing value requires validation	An untouched missing measurement may remain confirmed as Non renseigné . If the user supplies a replacement value, the correction shall require explicit validation before becoming a confirmed measurement.
BR-12	Safe noise is distinguished from unresolved information	Source metadata, interface fragments, timestamps, and other deterministically recognized non-business elements may be classified as safe noise. KPI-like or uncertain information shall remain unresolved unless sufficient evidence exists to classify it safely.
BR-13	Bulk ignore applies only to safe noise	A bulk ignore operation shall affect only source lines classified as safe noise. It shall never automatically discard unresolved KPI-like information.

Table 2.3 – continued

ID	Rule	Description
BR-14	Original source is preserved	The raw acquired content shall remain available after OCR extraction, normalization, association, validation, and source-line resolution so that a confirmed measurement can be traced back to its origin.
BR-15	New KPI creation is explicit	A new KPI definition shall be created only following an explicit user action. Before creation, the system shall verify that an equivalent definition does not already exist.
BR-16	Report confirmation requires resolution	A report shall not be confirmed while it still contains ambiguous, duplicate-review, corrected-missing, or unresolved KPI information requiring human intervention.
BR-17	Confirmed data drives analytics	Statistics, anomaly detection, trend analysis, forecasting, and maintenance-intelligence functions shall operate on validated historical measurements rather than directly on raw OCR or parser output.
BR-18	Missing values are excluded from numerical analysis	Missing information shall be excluded from numerical aggregates, statistical calculations, anomaly analysis, and forecasting unless a specific analytical method explicitly handles missing observations.
BR-19	Reports reflect validated information	Generated PDF and Excel documents shall be based on confirmed report data. Missing measurements shall remain identifiable as Non renseigné rather than being replaced by artificial values.

Table 2.3 – continued

ID	Rule	Description
BR-20	Reporting periods are deterministic	Weekly reports shall follow complete Monday-to-Sunday periods, while monthly reports shall represent complete calendar months. Automated generation shall use the corresponding completed reporting period rather than an arbitrary rolling interval.

These rules establish the semantic safeguards applied between raw acquisition and confirmed industrial information. They ensure that automation assists the maintenance engineer without silently changing the meaning of the original measurements.

The functional and quality requirements defined in the preceding sections, together with these business rules, provide the basis for the acceptance criteria presented in **Section VI**.

VI. Acceptance Criteria

The acceptance criteria translate the previously defined requirements into verifiable conditions. They provide a practical basis for determining whether the principal FactoryFlow workflows behave as expected.

TABLE 2.4 Main acceptance criteria for FactoryFlow

ID	Area	Acceptance criterion
AC-01	Acquisition	The user can acquire maintenance information through manual entry, pasted text, or an imported image. Image-based acquisition extracts visible text before the interpretation stage.
AC-02	Interpretation	Recognized KPI measurements are extracted while ambiguous values, missing measurements, duplicate observations, unresolved information, and source noise remain distinguishable.
AC-03	Missing values	An explicitly missing measurement remains represented as Non renseigné and is never automatically converted into zero. A real numerical zero remains a valid measurement.

Table 2.4 – continued

ID	Area	Acceptance criterion
AC-04	Human validation	Any measurement requiring human review provides an explicit resolution action before it can become part of the confirmed dataset.
AC-05	KPI association	An unresolved KPI can be associated with an existing indicator, while a genuinely new indicator can be created explicitly without forcing an unreliable fuzzy association .
AC-06	Duplicate observations	Multiple legitimate occurrences of the same KPI can be reviewed and validated independently without one occurrence silently overwriting another.
AC-07	Noise handling	Source metadata and safely identified non-business information can be ignored without automatically discarding unresolved KPI-like content.
AC-08	Traceability	A confirmed measurement remains traceable to its acquired source after interpretation, validation, persistence, and report generation.
AC-09	Draft management	An incomplete Review workflow can be saved as a draft and reopened later without losing its acquired information or current resolution states.
AC-10	Report confirmation	A report cannot be confirmed while information requiring mandatory human resolution remains unresolved.
AC-11	Historical analysis	Confirmed maintenance measurements can be consulted historically and used to generate statistical summaries and interactive trend visualizations. Missing values are excluded from numerical calculations.

Table 2.4 – continued

ID	Area	Acceptance criterion
AC-12	Maintenance Intelligence	Anomaly detection, trend analysis, forecasting, and contextual alerts use validated historical measurements rather than raw OCR or unreviewed parser output.
AC-13	Reporting	Confirmed maintenance information can be exported through structured PDF and Excel reports while preserving missing measurements as Non renseigné .
AC-14	Automation	Configured reporting schedules can trigger report generation and the corresponding notification or distribution operations according to the defined reporting period.
AC-15	Operational reliability	Temporary application or communication failures do not silently convert, discard, or confirm maintenance information without an explicit user or system resolution.

These acceptance criteria will also support the validation strategy presented later in the report, where the implemented workflows are evaluated against the expected system behaviour.

VII. Use Case Modeling

The requirements defined throughout this chapter were translated into a UML use case model in order to provide a global representation of the interactions between the user and FactoryFlow.

The platform has a single human actor: the **Maintenance Engineer**. This actor interacts with FactoryFlow throughout the maintenance-information lifecycle, from data acquisition and validation to historical consultation, reporting, automation, and maintenance-oriented analysis.

The principal interactions are represented in **Figure 2.1**.

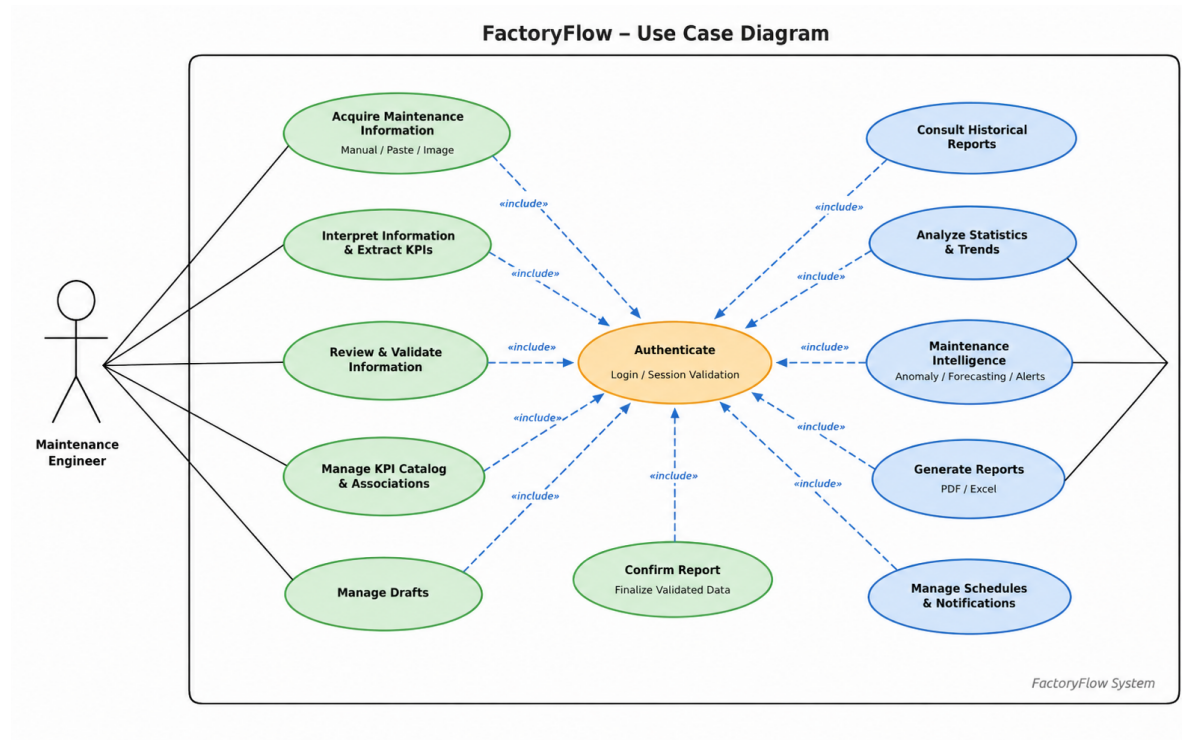


FIGURE 2.1 *FactoryFlow use case diagram*

As shown in **Figure 2.1**, authentication is a prerequisite for access to the protected functionalities of FactoryFlow. The corresponding use case is therefore included by the principal authenticated operations of the platform.

The Maintenance Engineer can acquire maintenance information through the supported acquisition mechanisms, review and validate interpreted data, manage KPI definitions and associations, preserve incomplete work as drafts, and confirm validated reports.

The same actor can subsequently consult historical information, analyse KPI statistics and trends, access Maintenance Intelligence results, generate reports, and manage the scheduling and notification functions associated with the reporting workflow.

The diagram therefore provides a consolidated representation of the functional perimeter previously established through the functional requirements, business rules, and acceptance criteria.

VIII. Conclusion

This chapter translated the industrial needs identified in **Chapter 1** into a structured specification for FactoryFlow.

The analysis identified the Maintenance Engineer as the principal human actor and defined the functional requirements covering acquisition, deterministic interpretation, human-controlled validation, KPI management, traceability, historical analysis, reporting, automation, and Maintenance Intelligence.

The non-functional requirements established the expected levels of data integrity, reliability, security, usability, maintainability, interoperability, and analytical reliability. These constraints were complemented by business rules that preserve the semantic meaning of the maintenance information, particularly the distinction between missing and zero values, the management of uncertainty, source traceability, and the use of validated data for analytical processing.

Finally, the acceptance criteria and UML use case model provided a verifiable representation of the expected system behaviour and its main user interactions.

Based on this specification, the next chapter presents the **software architecture and detailed design of FactoryFlow**, explaining how these requirements were translated into the technical components of the platform.

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- III. Backend Architecture**
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CHAPTER 4

FACTORYFLOW IMPLEMENTATION

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GENERAL CONCLUSION AND PERSPECTIVES

REFERENCES

CHAPTER A
ADDITIONAL TECHNICAL MATERIAL